

SRS

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1. Problem Statement

- There is a persistent problem of floor litter around indoor trash bins because users are unwilling or too lazy to dispose of trash properly.
- Human cleaners cannot operate 24/7, which leads to visible hygiene issues during peak hours.
- An intelligent robotic waste management system is required to autonomously intercept and collect trash to maintain cleanliness.

2. System Scope

- Indoor use only, on flat and obstacle-free floors.
- Operation is limited to a defined zone around fixed trash bins.
- The system does not handle liquid spills, fine dust, or stairs/ramps.
- The system does not handle large and heavy objects.

3. Functional Requirements

3.1 Vision-Guided Autonomous Waste Interception Robot (Catch Trash)

- The system shall detect and track airborne trash objects within the camera's field of view
- The system shall classify detected airborne objects by type and approximate size.

- The system shall track the trajectory of airborne objects in real time.
- The system shall predict the landing or interception position of the airborne object.
- The system shall navigate toward the predicted interception position before the object lands.
- The system shall perform the trajectory prediction within a bounded time window.
- The system shall update the object tracking at a minimum frequency.
- The system shall prioritise faster-moving objects over slower ones.
- The system shall dynamically adjust speed based on object distance and time-to-impact.
- The system shall stop or decelerate if interception becomes infeasible.
- The system shall detect failed interception attempts.
- The system shall hand over control to the Ground-Level Trash Collection Module upon interception failure.
- The system shall ignore objects below a configurable confidence threshold.
- The system shall operate only within predefined spatial boundaries.
- The system shall ignore objects detected outside the reachable interception zone.
- The system shall deactivate the interception function under low-light conditions below a defined threshold.
- The system shall log interception attempts and outcomes for performance analysis.
- The system shall support parameter tuning for interception thresholds.

3.2 Vision-Guided Floor Trash Collection Robotic Arm (Grab Trash)

- The system shall detect objects on the floor within the camera's field of view.

- The system shall detect trash objects on the floor within a 0.5 to 2-meter radius around the waste bin.
- The system shall provide coordinates for each detected trash object.
- The system shall classify detected floor objects by type and approximate size.
- The system shall achieve a minimum trash detection accuracy of 80% under normal indoor lighting.
- The system shall wait approximately 15 seconds before initiating trash collection for a newly detected object.
- The system shall prioritise pickup tasks based on detection confidence.
- The system shall allow the robotic arm to collect detected trash from the floor.
- The system shall execute collision-free movement to reach trash locations.
- The system shall grasp and transport collected trash to the bin.
- The system shall log performance data, including detection rate and pickup success rate.
- The system shall categorise operation failures for analysis (e.g., detection miss, localisation error, grasp failure).
- The system shall export performance metrics for review.

3.3 Shared

- The system shall independently control left and right wheel speeds to adjust the interception path.
- The system shall avoid collisions while navigating toward the interception point.
- The system shall limit the maximum speed to ensure safe operation in indoor environments.
- The system shall suspend interception if a human is detected within a safety radius.
- The system shall provide emergency stop functionality.

4. Non-Functional Requirements

4.1 Performance

- The system shall process visual input and update interception decisions within 200 ms per frame.
- The system shall maintain a minimum effective tracking rate of 15 frames per second.
- The system shall initiate motion within 100 ms after the interception point prediction.
- The system shall support real-time operation without requiring cloud computation.

4.2 Accuracy and Reliability

- The predicted interception point error shall not exceed ± 30 cm under controlled indoor conditions.
- The system shall correctly classify target trash objects with at least 80% confidence.
- The system shall perform consistently under normal indoor lighting conditions.

4.3 Safety

- The system shall limit maximum linear velocity to a safe indoor threshold.
- The system shall immediately halt motion upon detecting an obstacle within a predefined distance.
- The system shall not initiate interception if a human is detected within a safety radius.

4.4 Robustness and Fault Tolerance

- The system shall handle temporary loss of visual tracking without system failure.
- The system shall safely recover from failed interception attempts.

- The system shall continue operating if non-critical components fail.

4.5 Power and Resource Constraints

- The system shall operate continuously for at least X minutes on a single battery charge.
- The module shall not exceed the power budget allocated by the system power unit.
- CPU utilisation shall remain below 80% during normal operation.

4.6 Modularity and Maintainability

- The module shall be independently testable without the ground collection module.
- The module shall allow parameter adjustment without code recompilation.
- The module shall expose clear input/output interfaces for integration.

4.7 Usability and Configuration

- The system shall allow configuration of interception sensitivity parameters.
- The system shall provide visual or log-based feedback for debugging and evaluation.

5. System Decomposition

Vision-Guided Autonomous Waste Interception Robot

- Computer Vision + AI Prediction
- Movement Component
- Movement Control
- Sensor System (avoid objects & people)
- Schematics (hardware)
- Voice Recognition (additional)

Vision-Guided Floor Trash Collection Robotic Arm

- Computer Vision + AI classification
- Monitoring System
- Arm Component
- Arm Control
- Chassis Design (hardware)
- Stand Up (additional)

6. High-Level Architecture

7. Hardware Architecture

8. Software Architecture

```
robotic-waste-management/
|
|— docs/                # Documentation
|   |
|   |— ...
|
|— src/
|   |
|   |— main.py          # System entry point
|   |
|   |— config/          # Configuration (cross-cutting)
|   |   |
|   |   |— ...
|   |
|   |— events/          # Event infrastructure (cross-cutting)
|   |   |
|   |   |— __init__.py
|   |   |— event_bus.py
|   |   |— base_event.py
|   |   |— perception_events.py
|   |   |— motion_events.py
```

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├── safety_events.py
├── system_events.py

===== LAYER 1: PERCEPTION LAYER =====
├── perception/          # Layer 1: Sensing & Detection
│   ├── __init__.py
│   ├── camera_handler.py    # Camera 1 & 2 input
│   ├── vision_processor.py  # YOLOv8 detection
│   ├── airborne_tracker.py  # Track flying objects
│   ├── floor_detector.py    # Detect floor trash
│   └── models/
│       ├── yolov8_airborne.pt
│       └── yolov8_floor.pt

===== LAYER 2: DECISION LAYER =====
├── decision/           # Layer 2: Planning & Coordination
│   ├── __init__.py
│   ├── coordinator.py    # Main decision module
│   ├── trajectory_predictor.py # Predict landing point
│   ├── priority_manager.py # Task prioritization
│   └── dwell_timer.py    # 15s delay timer

===== LAYER 3: ACTION/MOTION LAYER =====
├── motion/             # Layer 3: Actuation & Control
│   ├── __init__.py
│   ├── mobile_base.py    # Module 1: Wheel control
│   ├── robotic_arm.py    # Module 2: Arm control
│   ├── path_planner.py   # Collision-free planning
│   └── kinematics.py     # IK solver

===== CROSS-CUTTING CONCERNS =====
├── safety/              # Safety (monitors all layers)
│   ├── __init__.py
│   ├── safety_monitor.py
│   ├── emergency_stop.py
│   └── sensors.py

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| | |
| | |— monitoring/          # Logging & Metrics (monitors all layers)
| | |   |— __init__.py
| | |   |— performance_logger.py
| | |   |— failure_classifier.py
| | |   |— metrics_exporter.py
| | |
| | |— utils/              # Common utilities
| | |   |— ...
| | |
|— tests/
|— data/
|— ...

```

9. Interface Definitions

Purpose: Decide *how* modules talk.

Examples:

- Vision → Decision: object position (x, y)
- Decision → Motor: speed (left, right)
- Decision → Arm: joint angles

If you skip this, integration will break.

10. Development Plan

- First Phase
 - Schematics (hardware)
 - Chassis Design (hardware)
 - Movement Component
 - Arm Component

- Sensor System (avoid objects & people)
- Second Phase
 - Computer Vision + AI Prediction (airborne)
 - Computer Vision + AI classification (floor)
 - Movement Control
 - Arm Control
- Third Phase
 - Monitoring System
 - Voice Recognition (additional)
 - Stand Up (additional)

11. Future Idea

- Throw trash at people to bully them
- Recharge the battery itself
- Refill or replace the trash bag autonomously.